



2022 Year 11 Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- Write your student number on this exam and multiple-choice answer sheet.
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks: 80

Section I – 15 marks (pages 2–5)

- Attempt Questions 1–15
- Write your answers on the answer sheet
- Allow about 25 minutes for this section

Section II – 65 marks (pages 6–22)

- Attempt Questions 16–20
- Allow about 1 hour and 35 minutes for this section

To be completed during the marking and checking process

Question	Multiple Choice	16	17	18	19	20	Total
Marks	15	14	22	10	9	10	80
Student's Score							

Student's comment:

Teacher's comment:

Disclaimer

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Section I

15 marks

Attempt Questions 1–15

Allow about 25 minutes for this section

Use the multiple-choice answer sheet for Questions 1–15.

- 1 Convert 75 000 *cm* to kilometres.
- A. 0.075 *km*
 - B. 0.75 *km*
 - C. 7.5 *km*
 - D. 75 *km*
- 2 What is the time difference between Sydney ($33^{\circ}\text{S}, 151^{\circ}\text{E}$) and Athens ($38^{\circ}\text{N}, 24^{\circ}\text{E}$)?
- A. 20*min*
 - B. 4*h* 44*min*
 - C. 8*h* 28*min*
 - D. 11*h* 40*min*
- 3 In the formula $K = \frac{G}{(1+p)^m}$, find K , correct to two significant figures, if $G = 80\,000$, $p = 0.05$ and $m = 6$.
- A. \$59 000
 - B. \$59 697.20
 - C. \$59 697.23
 - D. \$60 000
- 4 What is the amount of GST in the price of a laptop that has a total price of \$1320?
- A. \$120
 - B. \$132
 - C. \$1452
 - D. There is not enough information to find the solution

- 5 The width of a book was measured to be 14cm , correct to the nearest centimetre.

Calculate the percentage error, correct to two decimal places.

- A. 1.00%
- B. 3.57%
- C. 7.14%
- D. 14.00%

- 6 An example of a direct variation relationship is $y = \frac{1}{2}x$.

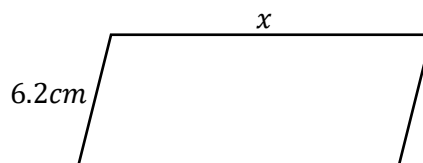
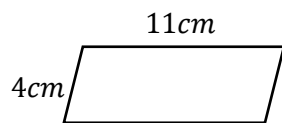
When graphed, what is produced?

- A. Straight line
- B. Parabola
- C. Cubic
- D. Hyperbola

- 7 What is the time 7 hours and 25 minutes after 6:56am on Monday 11 April, written in 24-hour time?

- A. 1229, Sunday 10 April
- B. 1421, Monday 11 April
- C. 1421, Tuesday 12 April
- D. 0221, Tuesday 12 April

- 8 The following parallelograms are similar.



NOT TO SCALE

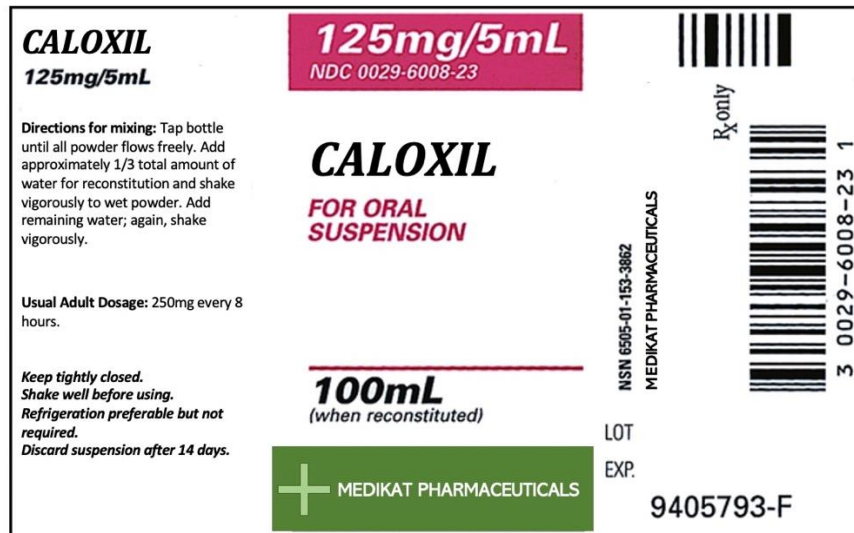
Calculate the unknown side, x .

- A. 1.55 cm
- B. 13.2 cm
- C. 17.05 cm
- D. 18.2 cm

- 9 Given Young's formula, calculate the approximate dosage of a 10-year-old child using the following medicine label below.

Young's Formula:

$$\text{Dosage for children 1 – 12 years} = \frac{\text{age of child (in years)} \times \text{adult dosage (mL)}}{\text{age of child (in years)} + 12}$$



- A. 2.3 mL
- B. 4.5 mL
- C. 56.8 mL
- D. 113.6 mL
- 10 State the gradient and y-intercept of the equation $y - 3x = 1$.
- A. gradient = 1 , y - intercept = -3
- B. gradient = -3 , y - intercept = 1
- C. gradient = 1 , y - intercept = 3
- D. gradient = 3 , y - intercept = 1
- 11 What is 4.1 tonnes equivalent to?
- A. 410 kg
- B. 41 000 kg
- C. 410 000 g
- D. 4 100 000 g

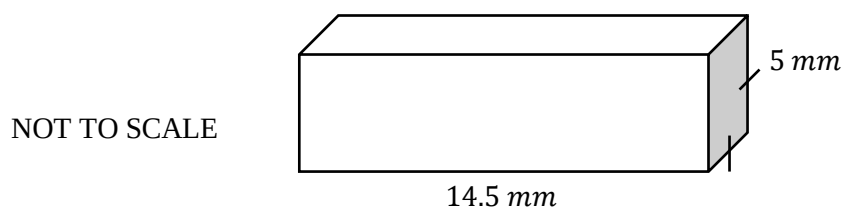
- 12 Use the straight-line method of depreciation to calculate the salvage value of a washing machine after 6 years, if it was originally purchased for \$980 and depreciates by \$155 per year.

A. \$50
B. \$155
C. \$825
D. \$930

- 13 Which type of car insurance covers damage to your vehicle in the case of an accident?

A. Compulsory third-party (CTP) insurance
B. Comprehensive insurance
C. Third-party property damage vehicle insurance
D. Accident insurance

- 14 Calculate the surface area of the rectangular prism.



A. 340 mm^2
B. 362.5 mm^2
C. 408 mm^2
D. 435 mm^2

- 15 Which is an example of stratified random sampling?

A. When a group of people volunteer to participate in a survey
B. Randomly selecting a sample of 10 students from a class of 30 by placing their names in a hat, mixing them and drawing their name
C. 15 items are selected from a population of 300 items by selecting every 20th item
D. A proportion of boys and girls are selected based on the gender ratio from a group of students

End of Section I

Year 11 Mathematics Standard

Section II Answer Booklet

65 marks

Attempt Questions 16–20

Allow about 1 hours and 35 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 16 (14 marks)

- (a) Make y the subject of $(2x + y)^3 = 8$. 2

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- (b) Sarah arrives at her friend's birthday party at 6:00pm and has her first standard drink. By 10:00pm she has had another 4 standard drinks and decides to stop so that she can drive home at midnight.

- (i) If Sarah weighs 75kg, calculate an approximate BAC using the following formula, correct to two decimal places: 2

$$BAC_{Female} = \frac{10N - 7.5H}{5.5M}$$

Where, N is the number of standard drinks consumed
 H is the number of hours of drinking, and
 M is the person's weight in kilograms.

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- (ii) Sarah is a P-plate driver and requires a BAC of zero. Will she be able to leave at midnight and drive home? Use the following formula to support your answer. 2

$$time = \frac{BAC}{0.015}$$

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Question 16 continues on next page

Question 16 (continued)

- (iii) Describe some limitations of this method of estimating Sarah's BAC. 2

[illegible]

- (c) The past few weeks has seen the cost of fuel in New South Wales increase dramatically. The average price of unleaded petrol has been \$2.00 a litre.

- (i) A car's fuel tank capacity is 48 litres. Calculate the cost of filling a full tank. 1

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- (ii) Graph the litres to cost. 2

This image shows a full page of blank graph paper. The grid consists of small, equal-sized squares formed by thin, dark gray lines. There are 20 columns and 20 rows of squares, creating a total of 400 square units. The background is white, and the grid covers the entire area of the page.

Question 16 continues on next page

Question 16 (continued)

- (iii) State the gradient and its significance in this graph. **2**

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- (iv) State one limitation of the graph. **1**

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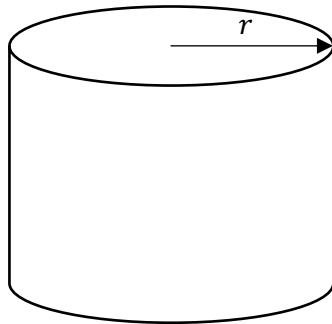
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End of Question 16

Question 17 (22 marks)

- (a) Peter was asked to calculate the volume of the cylinder below, by first measuring the radius of the circle.



NOT TO SCALE

- (i) He measured the radius to be 9.6 cm , correct to 1 decimal place. 1

Calculate the absolute error.

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- (ii) Find the limits of accuracy for the length of the radius. 2

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- (iii) Peter's calculations for finding the volume of the cylinder are below, where the exact height is given as 12 cm : 1

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi \times 9.6^2 \times 12 \\ &= 3474 \text{ cm}^3 \text{ (to the nearest whole)} \end{aligned}$$

State the capacity of the cylinder, giving your answer in litres.

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Question 17 continues on the next page

Question 17 (continued)

- (iv) Considering the volume given in part (iii), between what two values does the true volume of the cylinder lie? Answer to the nearest cubic centimetre. 2

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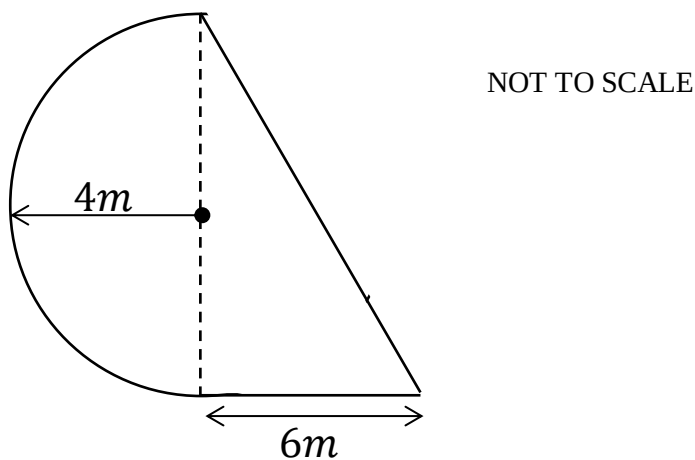
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Question 17 continues on next page

Question 17 (continued)

- (b) A piece of land, in the shape of a semicircle and right-angled triangle, has the following dimensions:



- (i) Calculate the perimeter of the land, to the nearest metre. 3

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- (ii) Pavers are to be laid across the entire piece of land, with each paver having an area of $0.5 m^2$ and costing \$12 per paver. 3

Calculate the cost of paving the entire piece of land.

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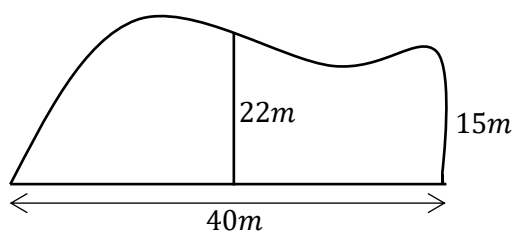
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Question 17 continues on the next page

Question 17 (continued)

(c)



3

Calculate the approximate area of the shape above, using two applications of the trapezoidal rule.

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(d)

The table below shows a list of foods, their serving sizes and the average number of kilojoules of energy in each serving.

3

FOOD	SERVING	ENERGY (kJ)
Nutrigrain	1 cup	525
Chicken Schnitzel	150g	1000
Egg (boiled)	1 medium	240
Hot chips	250g	1830
Grilled Octopus	2 cups	1170
Spaghetti Bolognese	375g	890
Watermelon	$\frac{1}{2}$ cup	410

Katrina consumes the following on Monday:

- 2 servings of Chicken Schnitzel
- 400g of hot chips
- 1.5 cups of watermelon

Using the table above, calculate Katrina's total energy intake, in kJ, on Monday.

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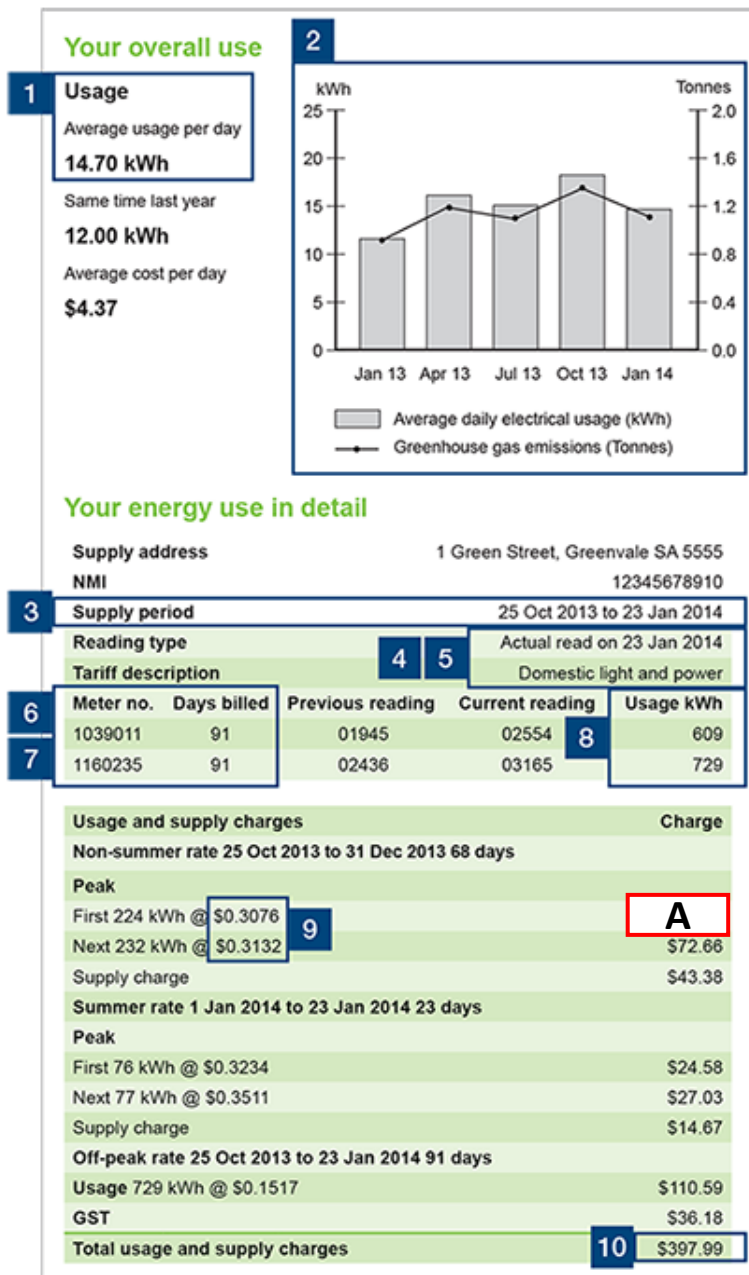
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Question 17 (continued)

(e) Below is an electricity bill.



Source: <https://www.sa.gov.au/topics/energy-and-environment/energy-bills/understanding-bills>

(i) How many days is the supply period? **1**

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(ii) What is the total amount of electricity used in kWh? **1**

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Question 17 continues on the next page

Question 17 (continued)

- (iii) What percentage of usage occurred in the off-peak period, correct to one decimal place? **1**

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- (iv) Calculate the value of **A**, as shown in the electricity bill on the previous page. **1**

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End of Question 17

Question 18 (10 marks)

- (a) George lives in Queensland and his friend, Joanne, lives in Sydney. He calls Joanne, and notices on his watch that it is 4:30pm. Accommodating for daylight savings and the call time which lasts 45 minutes, what time is it in Sydney when the phone call finishes? **2**

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Question 18 continues on the next page

Question 18 (continued)

- (b) Maria and Leni, who live in Sydney (UTC+10), decide to hire a private jet to celebrate New Year's Eve in style. They can celebrate New Year's Eve twice, due to the different time zones and travel time.

4

If they celebrate New Year's Eve in Sydney first, use the table below to state which other city would be possible to visit, in order to celebrate New Year's Eve twice. Support your answer with calculations.

CITY	UTC	Flight Time (hours)
Los Angeles	−7	12
Chile	−4	14.5
Madrid	+1	24
Auckland	+12	3

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Question 18 continues on next page

Question 18 (continued)

- (c) The timetable below is part of a light rail schedule.

	Monday to Friday: Departure Time					
Capitol Square	06:00	-	06:15	06:30	-	07:00
Paddy's Market	06:05	06:22	-	06:35	-	07:05
Convention Centre	06:09		06:24	06:39	07:00	07:09
The Star	06:19		-	06:49	07:10	07:19
Fish Market	06:24	X	06:39	06:54	07:15	07:24
Jubilee Park	06:27		06:42	06:57	07:18	07:27

- (i) How long does it take to get from Capitol Square to Jubilee Park? **1**

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- (ii) Assuming travel times are the same for all departures, find the time marked X. **1**

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- (iii) Sabrina's light rail was 8 minutes late, departing the Convention Centre at 06:48. It had made up 2 minutes by the time it arrived in Jubilee Park. At what time did it arrive? **2**

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End of Question 18

Question 19 (9 marks)

(a) Alice invested \$12 000 at 2.4% p.a. flat interest for 8 months.

- (i) Calculate the total amount of her investment. 2

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- (ii) After the 8 months, Alice's bank offers her a new rate for being a loyal customer. If she received \$365.76 in interest over the next 12 months, what was her new interest rate, correct to the nearest whole? 3

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Question 19 continues on the next page

Question 19 (continued)

- (b) The table below lists the maximum fortnightly payments for people receiving Youth Allowance.

Your circumstances	Your maximum fortnightly payment
Single, no children, younger than 18, and live at your parent's home	\$313.80
Single, no children, younger than 18, living away from your parent's home to study, train or look for work	\$530.40
Single, no children, 18 or older and live at your parent's home	\$367.00
Single, no children, 18 or older and need to live away from your parent's home	\$530.40
Single, with children	\$679.00
A couple, with no children	\$530.40
A couple, with children	\$577.40

Source: <https://www.servicesaustralia.gov.au/how-much-youth-allowance-for-students-and-apprentices-you-can-get?context=43916>

- (i) Diane is single, 21 years old and lives away from home. She has a two-year-old son. What is the maximum youth allowance Diane could receive? 1

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- (ii) Nick and Bree are a couple who are both 19 years old, with no children. Bree receives youth allowance, and their combined annual taxable income is \$49 302. If Bree's annual youth allowance is reduced by \$0.17 for every dollar their combined taxable income exceeds \$45 000, how much will Bree receive fortnightly in youth allowance payments? 3

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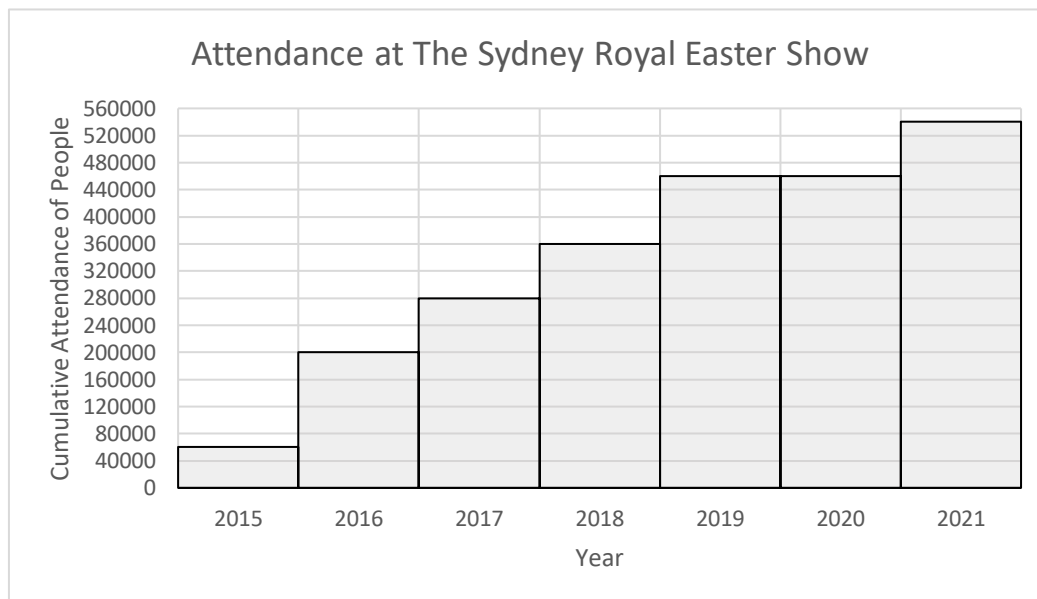
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End of Question 19

Question 20 (10 marks)

- (a) The radar chart shows the number of visitors who attended The Sydney Royal Easter show from 2015 to 2021.



- (i) In which year was the greatest attendance?

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- (ii) What could have accounted for the lack of attendance in 2020?

1

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Question 20 continues on next page

Question 20 (continued)

- (b) Jack has three sections in an exam. He has a 60% chance of passing Section I, a 34% chance of passing Section II, and a 90% chance of passing Section III.

- (i) How many possible outcomes are there? 1

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- (ii) Construct a probability tree diagram to represent this information, showing all possible outcomes. 3

- (iii) What is the probability of Jack receiving an overall pass, if he is required to pass at least two sections in the exam? 2

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- (c) Determine if there is an outlier in the data below. 2

4, 5, 6, 5, 4, 7, 2, 7, 12

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End of Section II

End of Examination

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

Mathematics Standard Reference Sheet

Measurement

Precision

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2} (x + y)$$

$$A = \frac{h}{2} (d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV (1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

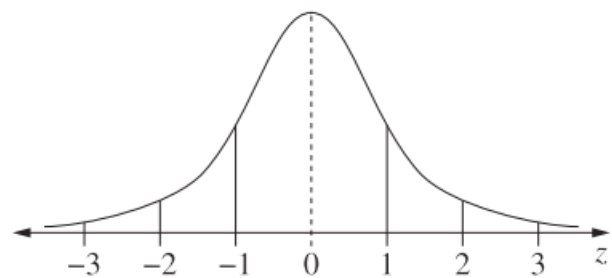
An outlier is a score

$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

MATHEMATICS STANDARD – MULTIPLE-CHOICE ANSWER SHEET**ATTEMPT ALL QUESTIONS**

Question	1	A ☐	B ☐	C ☐	D ☐
	2	A ☐	B ☐	C ☐	D ☐
	3	A ☐	B ☐	C ☐	D ☐
	4	A ☐	B ☐	C ☐	D ☐
	5	A ☐	B ☐	C ☐	D ☐
	6	A ☐	B ☐	C ☐	D ☐
	7	A ☐	B ☐	C ☐	D ☐
	8	A ☐	B ☐	C ☐	D ☐
	9	A ☐	B ☐	C ☐	D ☐
	10	A ☐	B ☐	C ☐	D ☐
	11	A ☐	B ☐	C ☐	D ☐
	12	A ☐	B ☐	C ☐	D ☐
	13	A ☐	B ☐	C ☐	D ☐
	14	A ☐	B ☐	C ☐	D ☐
	15	A ☐	B ☐	C ☐	D ☐

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